مقدار زمان برحسب ثانیه wait می‌کند

Wait_for Module

- wait for a set amount of time (default)
- Waiting for a port to become available
- wait for a file to be available or absent on the filesystem
- wait for active connections to be closed before continuing

```
- name: sleep for 300 seconds and continue with play
  wait_for: timeout=300
  delegate_to: localhost
```

مثلاً سرویس start کردیم نیاز داریم
به مدتی دستورات بعدی اجرا نشوند



Wait_for Module

برای یک پورت خاص Wait for

```
- name: Wait until the string "Jenkins is fully up and running" is in
the file catalina.out before continuing
  wait_for:
    path: /opt/apache-tomcat-9.0.0.M9/logs/catalina.out
    search_regex: 'Jenkins is fully up and running'
  tags: [wait_for_mod]

- name: After wait_for
  debug: msg="Jenkins is now up"
  tags: [wait_for_mod]
```




Exercise 19: wait_for module - wait for specific access to apache

1. Wait for a specific IP to access to your apache
2. If you have an access from the IP in the access_log file, then print a message
    1. Path: /var/log/httpd/access_log
    2. search_regex: 'ip_address'
3. Run the playbook and wait for the access
4. In the browser access the apache and check the playbook output



Exercise 19: (solved)

```
- name: ----- wait_for module -----
  wait_for:
    path: /var/log/httpd/access_log
    search_regex: '192.168.43.173'
  tags: [wait_apache]

- name: ----- wait_for module -----
  debug:
    msg: '192.168.43.173 access to apache'
  tags: [wait_apache]
```




Wait_for Module

```
- name: Wait 300 seconds for port 80 of the host to close active connections, don't start checking for 10 seconds
  wait_for:
    host: 172.20.30.40
    port: 80
    delay: 10
    state: started
```

Wait_for.state

absent, drained, present, started, stopped

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Exercise 20: wait_for module - be sure your service is up on a specific port

1. Config your apache to listen to port 8080
2. Stop apache service manually
3. Wait for port 8080 to be up on both remote machines and run the playbook
4. Set delay: 10
5. While you are waiting for port 8080, start apache on the remote machines
6. Check the output of the playbook



بعضی اوقات ممکن هست یه task به هر دلیلی fail شود
ممکن هست بعضی ماژول ها Return Value داشته باشد

Ignoring Failed Commands

- ✓ By default, playbooks will stop executing on a task fail. Sometimes, though, you want to continue on.
- ✓ The ignore_errors directive only works when the task is able to run and returns a value of '**failed**'. It does not make Ansible ignore undefined variable errors, connection failures, execution issues (for example, missing packages), or syntax errors.

بیوت دیش فرض Ansible اگه به ماژول fail بده Exit می‌کنه

```
- name: ignore errors
  command: /bin/false
  ignore_errors: yes
```

باست کردن این سوییچ اگه ماژولی fail شد بقیه task ها به کار خود ادامه می‌دهند
روی هر خطا این کار نمی‌کند



فرض کن ما یه دستور آپدیت زدیم روی host ها این task ها ممکنه خیلی طول بکشه تایم out ssh بشود باید یه سری task مارو هم زمان کنیم

Async actions

- ✓ In general you would want all your plays to be synchronous (do thing X, wait for it to be done/watch it, do thing Y).
- ✓ However, there are times when what you want to do will take a VERY long time or could kill your ssh connection off.

```
- name: update all packages
  yum:
    name: *
    state: latest
  async: 7500
  poll: 5
```

sec

→ interwall

هر ۵ ثانیه چک می کند این task انجام شده یا خیر

Async actions - concurrent tasks

- ✓ When poll is 0, Ansible will start the task and immediately move on to the next one without waiting for a result. (Fire 🔥 and forget!)

```
- name: ——— update all packages ———
  yum:
    name: *
    state: latest
  async: 7500
  Poll: 0
- name: ——— next task ———
  debug:
    msg: 'next task is running'
```

اگه ارتباط ما با host هامون از دست رفت از ماژول ← استفاده می کنیم

Wait_for_connection Module

✓ Waits for a total of timeout seconds for a connection

```
- name: wait_for_connection module
  wait_for_connection:
    timeout: 300
    delay: 10
    sleep: 5
```

Wait_for_connection.timeout

Maximum number of seconds to wait for

Wait_for_connection.delay

Number of seconds to wait before starting to poll

Wait_for_connection.sleep

Number of seconds to sleep between checks



Continue the playbook after restarting the host (reboot module)

✓ Reboot a machine, wait for it to go down, come back up, and respond to commands.

```
- name: --- reboot module ---
  reboot:
    reboot_timeout: 600
```




Mail Module

✓ sending emails from playbooks.

برای انجام یک کار به mail ها می‌فرستد

```
- mail:
    host: 172.20.*.*
    port: 25
    subject: Ansible-report
    body: Hello, This is a test :)
    from: sayahi@anisa.net (Sirous Sayahi)
    to: education<edu@anisa.net>
    attach: /etc/group /tmp/pavatar2.png
    headers: 'Reply-To=sayahi@anisa.net|X-Special="Something or other"'
    charset: utf8
```




Mattermost Module

✓ Sends notifications to your Mattermost server via the Incoming WebHook integration.

```
- name: - - - - - Send notification to mattermost - - - - -
  mattermost:
      url: https://chat.mycompany.ir
      api_key: ogiwjt5idjd14b3rtj8k8bop4c
      text: 'This is a test from {{ ansible_hostname }} :)'
      channel: devops-notification
      username: sayahi
  tags: mattermost_mod
```




Telegram Module

✓ Sends notifications via telegram bot

```
- name: - - - - - Send notification to mattermost - - - - -
  telegram:
      token: '1400058385:AAFGl0Wq8Z4YKSTYe2IEJGmVzO'
      chat_id: 000000
      msg: Ansible task finished
  tags: mattermost_mod
```




Subversion Module

```
- name: subversion module
  subversion:
    repo: http://svn.anisa.net/MyRepo/trunk
    dest: /srv/WD_Repo
    username: sayahi
    password: 123
    revision: 5
  tags: [svn_mod]
```




Git Module

✓ Manage *git* checkouts of repositories. The host that executes this module needs to have git>=1.7.1 .

```
- name: - - - - - git module - - - - -
  git:
      repo: 'http://anisa:anisa@192.168.56.35/git/hello-world'
      dest: /opt/hello
      revision: v1.0
  tags: [git_mod]
```




Get_url Module

برای وصل شدن به یک remote server و دانلود کردن یک فایل

- ✓ Downloads files from HTTP, HTTPS, or FTP to the remote server.

```
- name: Download file.conf
  get_url:
    url: ftp://download.anisa.co.ir/Course/LPIC-2/named.conf
    dest: /etc
    mode: 0440
    timeout: 30
    url_username: test
    url_password: test
```

کجا قرار بده؟

اگه login بخواد

- ✓ when run with --check, it will do a HEAD request to validate the URL but will not download the entire file or verify it against hashes.



Timezone Module

- ✓ configures the timezone setting

```
- name: set timezone to Tehran
  timezone:
    name: Asia/Tehran
```

- ✓ configures the ntp setting

```
- name: Configure ntp
  template: src=ntp.conf.j2 dest=/etc/ntp.conf
  notify:
    - restart ntp
```




Exercise 22: configure ntp and timezone

1. Install ntp on both remote hosts
2. Enable and start the ntpd service
3. Configure the ntp.conf and set the ntp servers as following: (use template module)

```
server 3.ir.pool.ntp.org
server 1.asia.pool.ntp.org
server 0.asia.pool.ntp.org
```

4. Restart the ntpd service
5. Set timezone to Asia/Tehran
6. Run timedatectl on the remote host to check the timezone



Synchronize Module

✓ Same as rsync command in your shell. rsync must be installed on both the local and remote host.

```
- name: synchronize of src to dest
  synchronize:
      mode: pull
      delete: yes
      archive: yes
      compress: yes
      rsync_path: sudo rsync
      src: "{{ src_dir }}"
      dest: "{{ dest_dir }}/{{ HN }}"
      rsync_opts:
       - "--exclude=workspace"
       - "--exclude=jobs/**/*.war"
       - "--exclude=jobs/**/*.jar"
       - "--exclude=jobs/**/*.ex
  tags: [Sync_Jenkins]
```

Mode

Specify the direction: push/pull

Delete

Delete files in dest that don't exist (after transfer, not before) in the src path.

Compress

Compress file data during the transfer.



iptables Module

✓ set up, maintain, and inspect the tables of IP packet filter rules in the Linux kernel.

```
- name: Block Specific IP
  iptables:
    chain: INPUT
    source: 172.20.30.40
    protocol: tcp
    destination_port: 22
    jump: DROP
    action: insert
    state: present
```

Iptables.chain

'INPUT', 'FORWARD', 'OUTPUT', 'PREROUTING', 'POSTROUTING', 'SECMARK', 'CONNSECMARK'

Iptables.action

'append', 'insert'

Iptables.jump

what to do if the packet matches it: DROP, ACCEPT



iptables Module

```
- iptables:
    table: nat
    chain: PREROUTING
    in_interface: eth1
    protocol: tcp
    match: tcp
    destination_port: 80
    jump: REDIRECT
    to_ports: 8080
    comment: Redirect web traffic to port 8080
```

Iptables.in_interface

Name of an interface via which a packet was received, use "!" to invert



firewalld Module

باز کردن فایروال بصورت داعمی

```
- firewalld:
    port: 80/tcp
    permanent: yes
    zone: public
    state: enabled
    immediate: true
  tags: firewalld_mod
```

Firewalls.service

Name of a service to add/ remove to/from firewalld -

Firewalls.immediate

Same as reload. Should this configuration be applied immediately, if set as

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Install Mariadb

```
- name: --- Install mariadb ---
  yum:
    name: mariadb-server
    state: installed
  tags: [Install_mariadb]

- name: --- enable and start mariadb service ---
  service:
    name: mariadb
    enabled: yes
    state: started
  tags: [start_mariadb]
```




Create a database in MySQL

✓ MySQL-python is needed by mysql_db module.

```
- name: --- Install MySQL-python ----
  yum:
    name: MySQL-python
    state: installed
  tags: [install_MySQL_python]

- name: --- create a database ---
  mysql_db:
    name: mydb
    state: present
  tags: [create_db]
```




Create a database in MySQL

✓ On target machine (Host_A)

```
# mysql
MariaDB [(none)]> show databases;
+--------------------------- +
| Database                   |
+--------------------------- +
| information_schema         |
| mydb                       |
| mysql                      |
| performance_schema         |
| test                       |
+--------------------------- +
5 rows in set (0.00 sec)
```




Exercise 25: install mysql and create DB

1. Install mariadb-server on both remote hosts
2. Enable and start the mariadb service
3. Install MySQL-python on both remote hosts
4. Create a database named db1 on both machines
5. Allow the mysql port (3306) on the machines to be accessible



Dump a database in MySQL

```
- name: --- dump a database ---
  mysql_db:
    name: mydb
    state: dump
    target: /home/mydb_dump.sql
  tags: [dump_db]
```




Restore a database in MySQL

```
- name: --- restore a database ---
  mysql_db:
    name: mydb
    state: import
    target: /home/mydb_dump.sql
  tags: [restore_db]
```




Add/Remove a user from a database

```
- name: --- Add mysql user ---
  mysql_user:
    name: anisa
    password: 123456
    priv: 'mydb.*:ALL'
    state: present
  tags: [add_mysql_user]
```


